Step 2 : Data

The given sample information are

Size of the Sample-1 (m) = 500

Size of the Sample-2 (n) = 800. Hence, both the samples are large.

Mean of Sample-1 ($\bar{x}$) = 58

Mean of Sample-2 ($\bar{y}$) = 57

Step 3 : Level of significance

$\alpha = 5\%$

Step 4 : Test statistic

The test statistic under the null hypothesis H_0 is

$$Z = \frac{\bar{X} - \bar{Y}}{\sigma\sqrt{\frac{1}{m} + \frac{1}{n}}}.$$

Since both m and n are large, the sampling distribution of Z under H_0 is the $N(0, 1)$ distribution.

Step 5 : Calculation of Test Statistic

The value of Z is calculated for the given sample information from

$$z_0 = \frac{\bar{x} - \bar{y}}{\sigma\sqrt{\frac{1}{m} + \frac{1}{n}}} \text{ as}$$

$$z_0 = \frac{58 - 57}{2\sqrt{\frac{1}{500} + \frac{1}{800}}}$$

$$z_0 = 8.77$$

Step-6 : Critical value

Since H_1 is a two-sided alternative hypothesis, the critical value at $\alpha = 0.05$ is $z_e = z_{0.025} = 1.96$.

Step-7 : Decision

Since H_1 is a two-sided alternative, elements of the critical region are defined by the rejection rule $|z_0| \geq z_e = z_{0.025}$. For the given sample information, $|z_0| = 8.77 > z_e = 1.96$. It indicates that the given sample contains sufficient evidence to reject H_0. Thus, it may be decided that H_0 is rejected. Therefore, the average performance of the students in the districts D_1 and D_2 in the national level talent search examination are significantly different. Thus the given samples are not drawn from identical populations.

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